

Professional Profile

Nirmal Nipuna Nanayakkara

Senior Front-End Engineer | UI/UX Specialist | Angular Expert | AI Application Developer

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I am a highly accomplished and passionate Senior Front-End Engineer and UI/UX Specialist with more than 12 years of professional experience designing, architecting, and delivering scalable enterprise-grade web and mobile applications. Throughout my career, I have consistently demonstrated the ability to transform complex business requirements into intuitive, user-centric digital experiences that drive measurable business outcomes.

My expertise spans the complete software development lifecycle, from product ideation and user research through architecture design, development, deployment, optimization, and maintenance. I have successfully contributed to and led the development of numerous enterprise solutions across industries including Logistics, Healthcare, Beauty, Food Distribution, E-Learning, Artificial Intelligence, Cognitive Computing, and Community Platforms.

Over the years, I have developed deep expertise in Front-End Engineering, User Interface Design, User Experience Design, Single Page Applications (SPAs), Micro-Frontend Architectures, Mobile Application Development, and AI-powered applications. My ability to combine technical excellence with design thinking allows me to create products that are not only visually appealing but also highly functional, scalable, and user-friendly.

Technical Leadership and Front-End Expertise

My primary specialization is modern front-end development using Angular, where I have extensive experience building complex enterprise applications from scratch. I have led front-end architecture decisions, established coding standards, designed reusable component libraries, and implemented scalable development practices that improve maintainability and team productivity.

I possess advanced knowledge of:

- Angular Framework
- TypeScript
- JavaScript (ES6+)
- HTML5
- CSS3
- SCSS
- RxJS
- NgRx
- Redux
- React
- Next.js
- Ionic Framework
- React Native
- Tailwind CSS
- Bootstrap
- Material UI
- GSAP Animations
- Three.js
- Chart.js

My experience includes designing and developing:

- Enterprise-grade web applications
- SaaS platforms
- AI-powered user interfaces
- Progressive Web Applications (PWAs)
- Mobile applications
- Single Page Applications (SPAs)
- Micro-Frontend ecosystems
- Component-based design systems

- Internal business platforms
- Data visualization dashboards
- Real-time applications

I am particularly skilled at creating maintainable front-end architectures that enable teams to scale applications efficiently while maintaining high performance and code quality.

User Experience and Product Design Excellence

My background in UI/UX development has provided me with a unique perspective that bridges the gap between design and engineering. I understand that successful products are not built solely through technical implementation but through a deep understanding of user behavior, business objectives, and design principles.

I have extensive experience in:

- User Interface Design
- User Experience Design
- Wireframing
- Interactive Prototyping
- Design Systems
- Information Architecture
- Responsive Design
- Mobile-First Design
- Accessibility Standards (WCAG)
- Design Thinking
- User Journey Mapping
- Interaction Design

Throughout my career, I have translated business requirements into wireframes, mockups, and production-ready interfaces that enhance user satisfaction while supporting business growth objectives.

Artificial Intelligence and Emerging Technologies

I am passionate about emerging technologies and have actively contributed to AI-driven projects and intelligent user experiences. My experience includes developing interfaces for AI applications, conducting research into next-generation human-computer interactions, and integrating AI-powered features into enterprise solutions.

At the Colombo Centre for Cognitive Computing, I worked on advanced AI interfaces and humanoid robot interaction systems, focusing on creating natural and engaging user experiences that improve communication between humans and intelligent systems.

I actively leverage modern AI tools including:

- Claude AI
- OpenAI
- Codex
- Microsoft AI Foundry

These technologies enable me to accelerate development workflows, generate technical documentation, improve code quality, assist with debugging, and create innovative user experiences.

Projects/ Applications Worked on

Throughout my professional journey, I have delivered projects and mission-critical applications serving thousands of users across multiple countries. Some notable projects are:

1. B&S Logistics Platform
2. Oracle MyOracle Intranet Platform
3. B&S Invoice Comparison Platform
4. CeyHallo Community Platform

B&S Logistics Platform

Technologies: Angular, TypeScript, RxJS, .NET, C#, SQL Server, REST APIs, JWT Authentication

As the lead Front-End Engineer and solution architect, I spearheaded the design, architecture, and development of the B&S Logistics Platform, a mission-critical enterprise application developed to modernize and centralize logistics operations across the organization.

The platform replaced multiple manual workflows and disconnected processes with a unified digital ecosystem, enabling real-time visibility, improved operational efficiency, and enhanced decision-making capabilities across logistics teams.

Key Responsibilities

- Led front-end architecture design and implementation using Angular and TypeScript.
- Collaborated closely with product owners, logistics stakeholders, and backend engineering teams to define scalable technical solutions.
- Designed and developed reusable UI components and modular application architecture to improve maintainability and scalability.
- Integrated multiple internal systems through RESTful APIs to establish seamless data exchange.
- Implemented advanced state management and reactive programming patterns using RxJS.

Major Features Delivered

- Real-time shipment tracking and monitoring
- Container-level logistics management
- Dynamic carrier rate comparison engine
- Logistics workflow automation
- Centralized logistics dashboard
- Order lifecycle tracking
- User role and permission management
- API integrations with B&S Photo and B&S Invoice applications

Business Impact

- Achieved 20% growth in customer acquisition.
- Reduced onboarding time by 30%.
- Improved order tracking accuracy by 50%.
- Significantly reduced manual operational effort.
- Improved visibility and transparency across logistics operations.
- Accelerated decision-making through real-time logistics insights.

B&S Invoice Comparison Platform

Technologies: Angular, TypeScript, .NET, C#, SQL Server, REST APIs, JWT Authentication

Contributed to the design and development of an intelligent invoice reconciliation system that automated the validation of supplier invoices against ERP system data.

The application was developed to eliminate manual invoice comparison processes, reduce financial discrepancies, and improve operational efficiency within finance and procurement departments.

Key Responsibilities

- Developed front-end modules using Angular and TypeScript.
- Collaborated with business analysts and finance stakeholders to translate requirements into technical solutions.
- Built advanced validation and reconciliation workflows.
- Designed data comparison interfaces that simplified invoice review processes.
- Optimized user workflows to minimize manual intervention.

Major Features Delivered

- Automated invoice validation engine
- ERP integration for order and pricing verification
- Quantity and pricing mismatch detection

- Approval workflow management
- Exception handling and reporting
- Audit trail and transaction tracking

Business Impact

- Significantly reduced manual invoice processing effort.
 - Improved invoice validation accuracy.
 - Enhanced financial data reliability.
 - Streamlined procurement and accounting operations.
 - Reduced processing time and operational costs.
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Oracle MyOracle Intranet Platform

Technologies: HTML5, SCSS, JavaScript, WCAG Accessibility Standards

Participated in the design and development of Oracle's enterprise intranet platform, delivering highly scalable and accessibility-compliant front-end solutions for a global workforce of more than 138,000 employees.

The project focused heavily on usability, accessibility, performance optimization, and reusable component architecture to support Oracle's internal digital ecosystem.

Key Responsibilities

- Designed and developed reusable front-end templates and UI components.
- Ensured full compliance with WCAG accessibility standards.
- Worked closely with Oracle implementation teams throughout development and deployment phases.
- Conducted front-end testing and quality assurance activities.
- Contributed to performance optimization initiatives.

Major Features Delivered

- Responsive enterprise portal interfaces
- Accessibility-compliant user experiences
- Reusable component framework
- Internal communication modules
- Employee self-service functionality
- Enterprise content management interfaces

Business Impact

- Improved accessibility and usability for over 138,000 employees worldwide.
 - Enhanced employee engagement and communication.
 - Delivered a scalable front-end architecture supporting future growth.
 - Improved operational efficiency across internal departments.
 - Maintained high standards of accessibility, performance, and reliability.
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CeyHallo Community Platform

Technologies: React Native, TypeScript, Redux, Tailwind CSS, Firebase, Firestore, Firebase Authentication, REST APIs

Designed and developed CeyHallo, a community-focused cross-platform mobile application that provides users with a centralized digital platform to connect, access community information, and engage with local activities and services.

The application was developed using modern mobile technologies and cloud-native backend services to ensure scalability, performance, and an exceptional user experience across both Android and iOS devices.

Key Responsibilities

- Led mobile application architecture and development.

- Designed reusable mobile UI components and shared service layers.
- Integrated Firebase Authentication and Firestore databases.
- Implemented state management using Redux.
- Developed REST API integrations and cloud-based services.
- Optimized mobile performance and user experience.

Major Features Delivered

- User registration and authentication
- Community news and announcements
- Real-time content updates
- Cloud-based data synchronization
- Cross-platform mobile experience
- Secure user profile management
- Community engagement features

Business Impact

- Enabled users to access community-related information through a centralized platform.
 - Improved communication and engagement within the community ecosystem.
 - Delivered a scalable mobile architecture supporting future feature expansion.
 - Enhanced application maintainability through reusable component design.
 - Provided a consistent user experience across Android and iOS platforms.
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Software Engineering Practices

I strongly believe that successful software development requires more than coding expertise. Throughout my career, I have consistently promoted engineering excellence through:

- Clean Code Principles
- SOLID Design Patterns
- Component-Based Architecture
- Code Reviews
- Technical Documentation
- Test-Driven Development
- Continuous Integration and Deployment
- Agile Methodologies
- Scrum Practices
- Performance Optimization
- Accessibility Compliance
- Security Best Practices

I have extensive experience collaborating with cross-functional teams consisting of product managers, UX designers, QA engineers, backend developers, DevOps engineers, and business stakeholders to ensure successful project delivery.

Cloud, DevOps, and Development Tools

I am proficient with modern development ecosystems and cloud technologies including:

Cloud Platforms

- Microsoft Azure
- Amazon Web Services (AWS)
- Firebase

Development Tools

- Git
- Bitbucket
- GitHub
- NPM

- Webpack
- Grunt
- Gulp

Project Management

- JIRA
- Agile Scrum Frameworks

Design Tools

- Figma
- Adobe XD

Backend & Integration Technologies

- Node.js
- Python
- .NET
- GraphQL
- REST APIs
- WebSockets
- MySQL
- Firebase Services

Leadership and Team Development

Beyond technical expertise, I am passionate about helping teams grow and succeed. Throughout my career, I have mentored junior developers, conducted knowledge-sharing sessions, performed code reviews, and established development standards that improve both individual and team performance.

I enjoy:

- Mentoring developers
- Conducting technical workshops
- Leading architecture discussions
- Reviewing code and enforcing best practices
- Sharing industry knowledge
- Supporting career development initiatives
- Building collaborative engineering cultures

I believe that knowledge sharing is one of the most important responsibilities of experienced engineers. Much of my own growth has been made possible through learning from talented professionals, and I strive to provide the same opportunities for others.

Professional Strengths

My key strengths include:

- Front-End Architecture Design
- Angular Enterprise Development
- UI/UX Design and Implementation
- SaaS Product Development
- AI Application Development
- Problem Solving and Critical Thinking
- Performance Optimization
- Technical Leadership
- Team Collaboration
- Agile Delivery
- Client Communication
- Stakeholder Management
- Mentorship and Coaching
- Attention to Detail
- Time Management

- Continuous Learning

Career Objective

I am seeking opportunities to contribute my expertise in front-end engineering, UI/UX design, SaaS product development, and AI-driven applications within innovative organizations that value technical excellence, creativity, and continuous improvement.

My goal is to leverage my extensive experience to build impactful digital products, lead high-performing engineering initiatives, and help organizations create exceptional user experiences that drive measurable business success while continuing to grow as a technology leader.

Education & Professional Certifications

Academic Qualifications

Bachelor of Science (Honours) in Computer Science

University College Dublin (UCD), Ireland
2010 – 2014

Successfully completed a comprehensive Computer Science degree program covering software engineering, programming methodologies, database systems, web technologies, computer architecture, software design principles, algorithms, data structures, and modern software development practices.

During the program, developed a strong foundation in:

- Software Engineering Principles
- Object-Oriented Programming
- Web Application Development
- Database Design and Management
- Computer Networks
- Human-Computer Interaction
- Software Architecture
- Systems Analysis and Design
- Data Structures and Algorithms
- Project Management
- User Experience Design

The degree provided both theoretical knowledge and practical industry-focused experience that became the foundation for a successful career in Front-End Engineering, UI/UX Design, Enterprise Application Development, and Artificial Intelligence-driven solutions.

Professional Certifications

Certified Software Engineer

HackerRank - hackerrank.com/certificates/be02598e6818

Successfully earned the Software Engineer Certification by demonstrating proficiency in:

- Software Development Best Practices
- Problem Solving
- Algorithms and Data Structures
- Software Architecture
- Clean Coding Principles
- Software Engineering Fundamentals

This certification validates advanced software engineering capabilities and practical coding expertise required to solve real-world engineering challenges.

Angular Certification

Udemy - [udemy.com/certificate/UC-50121f4e-4677-4003-a0eb-93518043fbae/](https://www.udemy.com/certificate/UC-50121f4e-4677-4003-a0eb-93518043fbae/)

Completed advanced Angular training focused on enterprise-level application development, including:

- Angular Architecture
- Components and Modules
- Dependency Injection
- Routing and Navigation
- Reactive Forms
- RxJS
- State Management
- REST API Integration
- Performance Optimization
- Enterprise Application Development

This certification further strengthened expertise in building scalable, maintainable, and high-performance Angular applications.

Angular Certification (Hard)

TestDome - testdome.com/certificates/aff0860ee6eb45d7a84c88472d42a421

Successfully passed the advanced Angular assessment designed to evaluate real-world Angular development skills.

Key areas assessed included:

- Angular Framework Architecture
- TypeScript Development
- Reactive Programming
- Component Design
- Dependency Injection
- Application Optimization
- Performance Best Practices
- Enterprise Front-End Development

This certification demonstrates strong technical competency and practical expertise in modern Angular application development.

Continuous Professional Development

As a technology professional, I strongly believe in continuous learning and staying ahead of industry trends. I actively invest time in expanding my knowledge across:

- Artificial Intelligence
- Generative AI Technologies
- Front-End Architecture
- Cloud Computing
- Modern JavaScript Ecosystem
- UX Research & Design Systems
- Performance Engineering
- Accessibility Standards (WCAG)
- Mobile Application Development
- Enterprise Software Architecture

I regularly leverage industry-leading platforms, technical communities, conferences, documentation, and AI-assisted development tools to remain at the forefront of emerging technologies and software engineering best practices.

Professional Philosophy

Education extends beyond formal qualifications and certifications. Throughout my career, I have maintained a strong commitment to self-learning, experimentation, mentorship, and knowledge sharing. This growth mindset has enabled me to evolve from a UI/UX Developer into a Senior Front-End Engineer capable of leading enterprise-scale projects, mentoring teams, architecting complex solutions, and contributing to AI-powered products used across global organizations.